HospitAI

A hospital data analytics platform that combines
patient grouping, clinical sequence mining,
and AI-generated interpretation

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Overview

In short, **HospitAI** is an **end-to-end** analytics pipeline that ingests raw EHR data, applies **unsupervised patient clustering** and **clinical sequence mining**, and provides **AI-generated** interpretations to support data-driven hospital insights.

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Introduction and context

Context and challenges (EHR data)

- EHR (Electronic Health Record) systems gather huge volumes of clinical data, but most of it stays **unused**. Also, querying it requires SQL expertise that most medical staff lack.
- A global analysis on that data hides **patient diversity**: an acute case is very different from the many conditions of an elderly patient.
- Medical care is strongly **time-based**. The order of actions matters more than how often they happen.
- Classic algorithms return lists of abstract codes that clinicians cannot interpret quickly.

HospitAI addresses these challenges with: patient grouping (clustering), clinical sequence mining, an AI interpretation layer, and an AI Chat Agent.

Main objective and sub-objectives

Main objective

Build the **end-to-end** **HospitAI** pipeline that turns raw clinical data into explainable medical decisions.

Sub-objectives

1. **Patient grouping:** cluster patients into homogeneous groups using K-Prototypes on mixed-type data.
2. **Sequence mining:** extract and rank treatment patterns over time using PrefixSpan and GML score.
3. **AI verbalization:** generate readable clinical summaries per cluster via an LLM verbalizer.
4. **SQL agent:** allow patients or medical staff to query clinical data in plain English via a LangGraph agent; also responds to general health-related questions.

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Related work

Related work: what others did

Category	Papers	What they do
Surveys	Rosa (2022), Flothow (2023)	Rosa distinguishes sequence mining from process mining in pathway analysis. Flothow separates full-trajectory clustering from subsequence pattern mining.
Clustering and mining	Najjar (2018), Baudrier (2022), Vogt (2017)	Najjar clusters individual visits with K-Prototypes then derives pathways. Baudrier links trajectory clustering to outcomes in kidney cancer. Vogt clusters heart-failure pathways from insurance claims.
Sequence mining	Perer (2015)	Mines frequent event sequences from EMRs and links them to patient outcomes via visual analytics.
Patient clustering	Yang (2024), Aljohani (2024)	Yang clusters EHR patients to reveal health risks. Aljohani evaluates eight clustering methods for patient stratification.
Process mining	Murazzano (2026)	Applies process mining to chronic respiratory patients, observing high variability in care processes.

Related work: gaps and our contribution

Category	Gap
Surveys	Overview only; no unified implementation
Clustering and mining	Visit-level grouping; single-disease scope; steps applied in a different order
Sequence mining	Frequency only; no clinical ranking metric to filter trivial patterns
Patient clustering	No sequential analysis component
Process mining	No AI interpretation or verbalization layer

What HospitAI adds

- **Patient-level** mixed clustering with K-Prototypes across **22,284** patients, not visit-level
- Cross-domain applicability, not limited to a single disease or patient type
- **Clinically ranked** sequences via the GML score, which rewards genuine statistical association beyond raw frequency
- **LLM verbalization** that generates readable cluster summaries for both clinicians and patients
- **Natural-language SQL agent** built on LangGraph, enabling clinical queries without SQL expertise

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The HospitAI analytics pipeline

Pipeline architecture and ETL (Stage 1)

- The architecture is split in two: an offline analysis pipeline in **Python** that processes raw EHR data and produces clustering and sequence mining results, and a fast web application (**Next.js and FastAPI**) that displays those results and allows medical staff to perform CRUD operations on the data.
- **Stage 1 - ETL (extract, transform and load):** Synthea complex synthetic EHR dataset (**22,284** patients), used to ensure full **privacy compliance** (GDPR). Missing values filled by **mode / median**.

figure1.png

Fig. 1. The HospitAI pipeline

Patient Grouping (Stage 2)

- **Why K-Prototypes:** it handles mixed data (numeric and categorical) natively. K-Means fails on categorical attributes; K-Modes fails on numeric ones.
- Many features were **derived** from raw EHR data rather than read directly. For example, the **clinical severity score** is computed from condition count, medication load, and procedure frequency.
- **K=4** was selected via an **elbow plot** on the Gower inertia curve, which showed a clear inflection point at 4 clusters.
- At cluster creation, the **AI layer** receives the feature values via prompt and generates a **title**, **insights**, and **recommendations** tailored to each patient group.

Huang mixed distance (γ = categorical weight)

$$d(X,Y) = \sum_{j=1}^{p}(x_j - y_j)^2 + \gamma \sum_{j=p+1}^{m} \delta(x_j, y_j)$$

Outlier detection

$$z = \frac{x - \mu}{\sigma} > 6 \;\Rightarrow\; \text{class } (-1)$$

HospitAI - Patient Clusteri

Sequence Mining (Stage 3)

- Patient timelines are built over a **2-year window**; consecutive duplicates removed to avoid noise.
- **PrefixSpan** extracts frequent subsequences with minimum support **1%**.
- **Why not frequency alone:** common events carry no clinical signal. The **GML score** measures genuine statistical association between events.
- A **diversity bonus** rewards patterns mixing medications, procedures, and observations.
- Mining runs at **two levels**: globally and per cluster.

GML score

$$\mathrm{GML}(p) = \frac{s(p)/n}{\left(\prod_{i \in p} s(i)/n\right)^{1/k}}$$

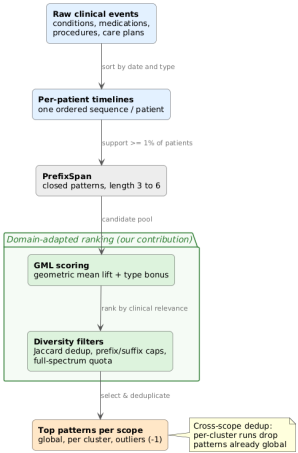


Fig. 2. The sequence mining pipeline

HospitAI - Patient Cluste

The AI layer and the SQL agent (Stage 4)

AI Verbalization

- Clusters are interpreted **automatically at creation**: the LLM generates a **title**, **description**, **insights**, and **recommendations** from precomputed feature values.
- Interpretation can also be **triggered on demand**.
- Two output modes: **medical-oriented** (clinicians) and **patient-oriented** (plain language).
- **Sequences** can also be verbalized by the AI layer.

SQL Agent

- Translates natural language to SQL; behavior adapts to **user role**.
- Answers general medical questions with a disclaimer: responses are **not a substitute for a doctor**.
- Out-of-scope questions are **refused**.

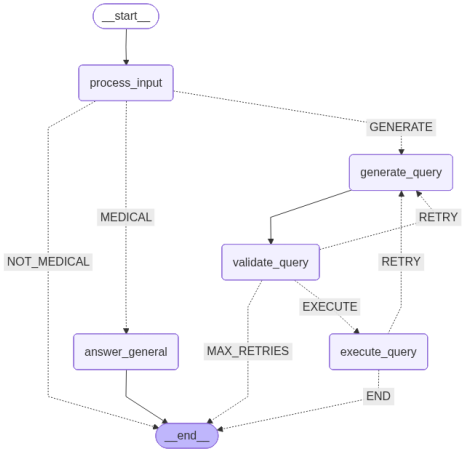


Fig. 3. The agent state machine (LangGraph)

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Simulations and experimental results

Simulation: experimental setup and data

- After ETL, in Stage 2, most features that will be used for clustering are extracted through **SQL aggregation queries** over the raw EHR tables.
- Some are **derived** from others via feature engineering, e.g., tabulating clinical events into scores or ratios. Below there are listed all the features.

Group	Features
Demographics	age, gender, race, ethnicity, marital status
Encounter utilization	inpatient, outpatient, emergency, wellness, ambulatory, urgentcare, SNF, hospice counts; dominant encounter class; total encounter duration (hours); days since last encounter; routine care count, acute care count, severe care count *(derived)*; preventive-to-reactive ratio *(derived)*; out-of-hours flyer; has inpatient history; readmission risk tier
Conditions	total conditions, active conditions count, condition burden *(derived: low/medium/high)*; has_diabetes, has_hypertension, has_heart_disease, has_cancer, has_kidney_disease, has_mental_health, respiratory disease profile
Medications and procedures	total medications, lifetime medication days, is_polypharmacy, medication intensity *(derived: low/medium/high)*, pain management tier; total procedures; clinical severity score *(PCA-derived from procedures, medications, and active conditions)*
Vitals and observations	avg BMI, BMI type *(derived: Underweight to Obese III)*; avg systolic BP, BP risk category *(derived)*; avg glucose, glucose risk category *(derived: Normal/Prediabetic/Diabetic)*; QALY, DALY
Behavioral and risk	substance use profile, smoking status, cholesterol risk category, pregnancy history; distinct providers seen; dominant care plan

Results: the obtained clusters

- **Cluster 0 (Young Injury):** young patients, acute trauma, emergency and urgentcare encounters, low chronic burden.
- **Cluster 1 (Maternity):** predominantly female, maternal care plans, low medication load, wellness encounters.
- **Cluster 2 (Older High Risk):** elderly, polypharmacy, hypertension, diabetes, heart disease, long encounter duration.
- **Cluster 3 (Pediatrics):** children, minimal active conditions, low procedure counts, wellness dominant.
- **Cluster (-1) (Outliers):** extreme cases (oncology, dialysis, terminal care), stored separately for independent analysis.

Why the clusters are good

- Each cluster maps to a **recognizable patient type** that a clinician can immediately interpret.
- The separation is driven by **clinical behavior**: conditions, medications, and encounter patterns, not superficial demographics.
- Each group has a **distinct care trajectory** that justifies different treatment and prevention strategies.

Results: cluster profiles

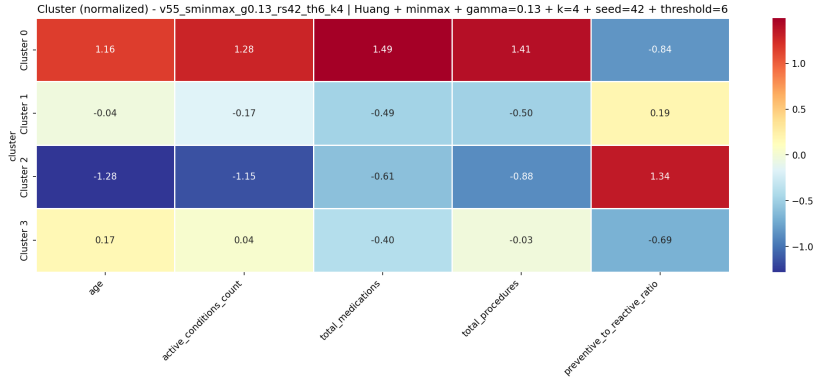


Fig. 4. The obtained cluster profiles - fully AI interpreted

Results: meaningful clinical sequences

PrefixSpan (min. support 1%), sequences ranked by **GML score**. Top non-dental patterns, global and per cluster:

Global	Laceration → Wound care → Suture open wound	0.97
Global	Fluticasone/Salmeterol inhaler → Albuterol → Spirometry	0.96
C0 (Young Injury)	Laceration → Wound care → Suture open wound	0.98
C0 (Young Injury)	Concussion → Head injury rehabilitation	0.75
C1 (Maternity)	Tubal ligation → Minor surgery → Postoperative care	0.91
C1 (Maternity)	Normal pregnancy → Pregnancy test → Fetal ultrasound	0.73
C2 (Older High Risk)	Fluticasone/Salmeterol inhaler → Albuterol → Spirometry	0.95
C2 (Older High Risk)	Epoetin Alfa → Renal dialysis → Epoetin Alfa	0.82
C3 (Pediatrics)	Allergic reaction → Epinephrine injection → Prednisone	**1.00**
C3 (Pediatrics)	Seizure disorder → EEG monitoring	0.91
C3 (Pediatrics)	Otitis media → Acetaminophen → Amoxicillin	0.45
C(−1) (Outliers)	Renal medication review → Tacrolimus → Kidney transplant	0.97
C(−1) (Outliers)	Cisplatin → Chemo+radiation → Hospice care	0.85

Table 1: Most meaningful sequences found per cluster (ranked by GML score)

Results: cluster interpretation and Chat Agent

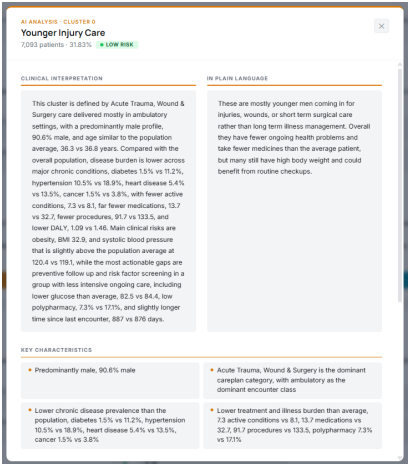


Fig. 5. AI-generated cluster interpretation

chatbot.png

Fig. 6. Chat Agent - example interaction

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Discussion and performance review

Discussion: benchmark logic and results validation

- **629** total runs across **60** feature configurations (v1–v60). Each config defines a distinct set of numerical and categorical features.
- Per config, a **nested loop** varied: $k \in \{3, 4, 5, 6\}$ and Z-score threshold $\in \{4, 6\}$. For the best candidates, gamma $\in \{\text{auto}, 0.07, 0.13, 0.17\}$ and random seed $\in \{7, 42, 100\}$ were also tested.
- Each run was evaluated with three metrics: **Gower Silhouette**, **Calinski–Harabasz**, and **Davies–Bouldin**, combined via a min-max normalized empirical score. Runs that split patients only by gender were excluded.
- Runs were sorted by score; the best configuration per $k \in \{4, 5, 6\}$ was selected (see table).
- For **sequence mining**: patterns ranked by **GML** (geometric mean lift). A **diversity bonus** $(1 + 0.3\,(t - 1))$ rewards patterns spanning multiple event types.

Optimization score (min-max normalized, empiric)

$$\text{Score} = 0.5\,\text{Gower} + 0.3\,CH_n + 0.2\,DB_{inv}$$

Config	k	Gower	Score
v55	**4**	**0.350**	**0.82**
v40	5	0.324	0.74
v60	6	0.287	0.70

v55 feature set ($k = 4$)

Numerical (5)	age, active conditions, total medications, total procedure prev./reactive ratio
Categorical (8)	gender, pregnancy history, h disease, diabetes, hyperten condition burden, BMI type, dominant care plan

Table 2: v55 feature set. Results above are based on these features.

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Conclusions and future work

Conclusions and future work

Contribution

HospitAI shows that **joining** mixed clustering with ranked mining turns the noise in EHR data into **clear medical paths**.

What the results enable

- **Clusters** support risk stratification and targeted resource allocation per patient group.
- **Sequences** can seed or validate clinical treatment protocols and guidelines.

Limitations

- **Synthetic** data (Synthea).
- It relies on an external language model, which raises hospital governance concerns.

Future work

- Direct clinical review with specialist doctors.
- Testing on **real**, anonymized databases.

References

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Q & A

Thank you for your attention!

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Appendix: formulas

Appendix I: clustering and ETL pipeline

K-Prototypes distance (Huang)

$$d(X,Y) = \sum_{j=1}^{p}(x_j - y_j)^2 + \gamma \sum_{j=p+1}^{m} \delta(x_j, y_j), \qquad \delta(x_j, y_j) = \begin{cases} 0, & x_j = y_j \\ 1, & x_j \neq y_j \end{cases}$$

Outlier detection (Z-score, post-log)

$$z = \frac{x - \mu}{\sigma} \;\Rightarrow\; z > 6 \Rightarrow \text{class } (-1)$$

Scaling and normalization

$$X_{norm} = \frac{\ln(X) - \min(\ln X)}{\max(\ln X) - \min(\ln X)}$$

Appendix II: sequence mining and evaluation

Geometric Mean Lift (GML), base form

$$\mathrm{GML}(p) = \frac{s(p)/n}{\left(\prod_{i \in p} s(i)/n\right)^{1/k}}$$

Final GML (with a diversity bonus, t = number of event types)

$$\mathrm{GML}_{final}(p) = \mathrm{GML}(p) \cdot \left(1 + 0.3\,(t-1)\right)$$

Weighted linear optimization (hyperparameter tuning)

$$\mathrm{Score}_{config} = 0.5 \cdot \mathrm{Gower}_{norm} + 0.3 \cdot CH_{norm} + 0.2 \cdot DB_{inv_norm}$$